

**Final Project Report: Avoiding Traffic Congestion in Toronto Centre using Least Cost
Path Analysis (LCPA)**

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Abstract

Traffic congestion is a major issue in urban areas, leading to longer travel times and increased fuel consumption. This study focuses on improving route planning within the Toronto Centre Ward by applying Least Cost Path Analysis (LCPA) to identify more efficient travel routes. Our approach integrates real-time congestion data, speed limits, and the location of traffic signals to create a realistic model of travel conditions. Additionally, we forecasted congestion levels for the following day to explore how predictive modeling can be used to enhance route optimization.

The analysis revealed that incorporating real-time and forecasted congestion data into LCPA can effectively reduce travel time by identifying routes that avoid high-congestion areas. The results demonstrated that accounting for traffic conditions, both current and expected, allows for more accurate and practical route planning in urban settings. However, the study also encountered some limitations, such as challenges in handling missing data and the need for more detailed traffic information to improve the precision of the cost metric.

Despite these limitations, the findings provide a practical approach to improving urban route planning. Future work should focus on refining the cost metric to better account for the complexities of traffic patterns and integrating more detailed and dynamic data sources. Expanding the application of this methodology to larger road networks could lead to further improvements in traffic management and provide better route planning that could help mitigate the effects of urban traffic congestion.

Introduction

Traffic congestion has become a growing concern in many cities around the world, particularly in densely populated urban areas. As the number of vehicles on the road continues to increase, existing infrastructure often struggles to accommodate the demand, leading to frequent bottlenecks and slowdowns. This congestion not only results in longer commute times for drivers but also contributes to environmental pollution due to idling vehicles and inefficient fuel use. Additionally, the constant stop-and-go nature of congested traffic can lead to higher rates of vehicle wear and tear, increasing maintenance costs for drivers. The impact extends beyond individual drivers, as businesses experience delays in goods transportation, and emergency services find it difficult to respond quickly to incidents. Research by Wang and Zhang (2019) has shown that the density of human activities and the layout of urban areas significantly impact daily traffic conditions, leading to longer travel times and more frequent congestion. This complex relationship contributes to the persistent congestion seen in major urban centers.

Toronto has recently been recognized as the most congested city in North America, with traffic delays surpassing those in cities like New York and Los Angeles (TomTom, 2024). The city's downtown core, particularly the Toronto Centre Ward, experiences severe congestion, making it a focal point for traffic management studies. Previous research into Toronto's commuter patterns has provided valuable insights into the factors influencing travel distances within the city's commuter shed (Axisa et al., 2012). However, these studies have not fully explored the potential of integrating predictive modeling with real-time geospatial data to optimize traffic flow in a targeted urban area.

While commercial platforms like Google Maps and Apple Maps offer real-time traffic updates and route optimization, this project aims to implement a practical application of Least Cost

Path Analysis (LCPA) to study how such methods can be utilized to avoid congested areas and save travel time.

The primary objective of this project is to use LCPA to identify the most efficient travel routes within downtown Toronto by considering traffic congestion, speed limits, and traffic signal locations as key factors. Additionally, the study will forecast congestion levels for the following day, providing a proactive approach to route optimization. This research aims to validate the use of LCPA in a real-world setting and contribute to the development of more tailored and data-driven traffic management solutions that could complement existing commercial platforms.

Data Collection and Preprocessing

The data used for this project includes the data required to create a cost of travel to determine the most efficient path in terms of time. We decided on using the traffic congestion as the primary Before collecting the data, it is essential to identify the target area of analysis and define a bounding box such that only the desired data is collected. Since our focus was the Toronto Centre Ward, the bounding box containing this ward was defined. Then, we queried the boundary of the Toronto Centre ward from OpenStreetMap. All the data collected was subsetting to fall within this boundary. In the following section, we discuss the data collected and the significance of the collected data for this project.

Road Network Data:

The Road Network Data used in this project was sourced from OpenStreetMap. This is a vector data containing line strings that represent the road segments forming the road network in Toronto Centre ward. This vector data serves as the backbone for the Least Cost Path Analysis (LCPA) conducted in this study. Additionally, this data also includes a column called "Maxspeed," which contains associated speed limits of the road segments. The speed limits are important as

they will allow us to calculate the time required to travel from point A to B while complying with the legal speed constraints.



Figure 1: Toronto Centre Ward Road Network

Traffic Congestion Data:

The traffic congestion data is the core of this project as we consider congestion level on the road to be the primary cost in our analysis. The data was extracted using the `GoogleTraffic` package in R, which communicates with the Google Maps API to extract the congestion levels of the roads in real-time and returns a raster layer with discrete congestion values along the road segments. The congestion values range from 1 to 4 with 1 representing free traffic flow or no congestion and 4 representing heavy congestion. One thing to be noted here is that the function used to query this congestion data requires a boundary polygon as an input. We used the boundary

of the Toronto Centre ward for this purpose, so the queried data only includes congestion values for roads within that boundary.

Since one of the project's objectives was to forecast the congestion values for the next day such that LCPA could be performed on the forecasted congestion values, we collected the congestion data by running an automated script that ran for seven days, collecting congestion data at different times throughout the day.



Figure 2: Traffic Congestion Level in Toronto Center Ward

Traffic Signal Data:

The Traffic Signal Data was sourced from the Toronto Open Data Portal and was subsetting to contain the locations of traffic signals present in Toronto Centre Ward. Traffic signals can significantly impact travel times, especially in densely populated areas like downtown Toronto, where frequent stops can cause substantial delays. By incorporating this data into the Least Cost Path Analysis (LCPA), we can account for the delays induced by traffic signals more accurately.

As a result, our routing system not only considers congestion levels but also effectively accounts for time lost at intersections which will ultimately reduce the overall travel time.

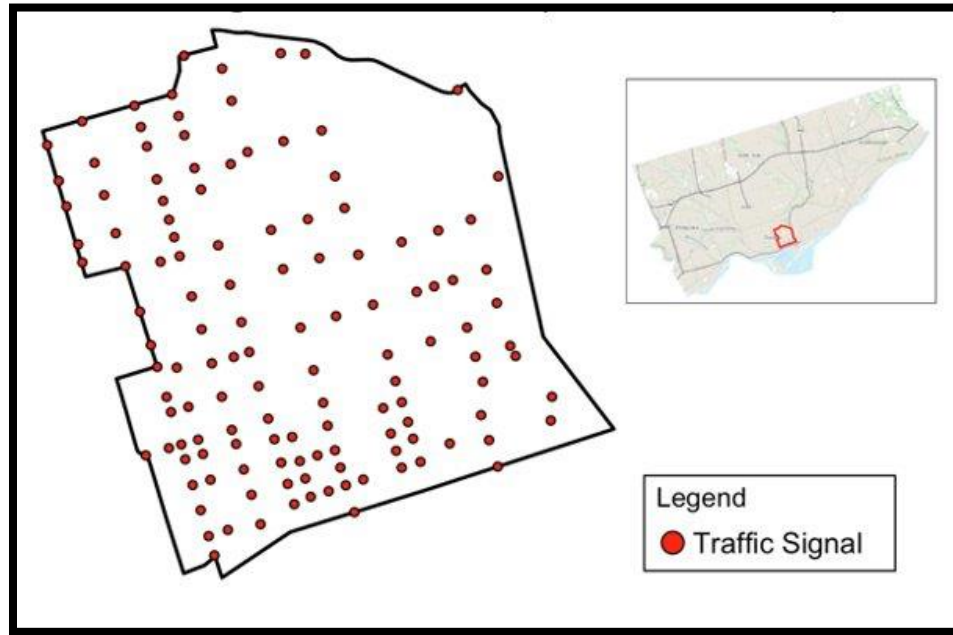


Figure 3: Traffic signal locations in Toronto central ward

Analytical Methods

In this section we will discuss the methods used in our analysis in detail. The primary objective of the study was to identify the most efficient travel routes within the Toronto Centre Ward by applying Least Cost Path Analysis (LCPA). The methods employed in this analysis include formulating a cost metric based on travel time, constructing a detailed graph of the road network, and applying pathfinding algorithms to determine the optimal routes. We further extend the analysis by forecasting the congestion values and perform LCPA using the forecasted congestion values to get an efficient path for the next day.

Formulating the Cost Metric

In Least Cost Path Analysis (LCPA), the "cost" represents the effort or resources required to travel from one point to another. The accuracy of the LCPA heavily depends on how well this cost reflects real-world conditions. In our analysis, path efficiency is based on travel time, so it was very important to develop a cost metric that accurately represents the time required to travel along each road segment.

To achieve this, we formulated a cost metric based on travel time. The basic principle is that higher congestion, lower speed limits, and a greater number of traffic signals will result in longer travel times. Therefore, the cost metric must account for these factors to effectively reflect the time taken to travel between points.

Simply put, the time taken to go from one point to another is calculated as:

$$Time = \frac{Distance}{Speed} \dots \dots \dots (eqn. 1)$$

In our case, assuming all drivers adhere to the speed limit, the travel time on a road segment would be:

$$Time = \frac{Segment Length}{Speed Limit} \dots \dots \dots (eqn. 2)$$

Here, the "Segment Length" refers to the distance of the road segment (or edge length in the graph). However, since we also need to account for the effects of congestion and potential delays due to traffic signals, we extended this formula to include these factors:

$$Weight (travel time) = \frac{Segment Length}{\left(Speed Limit \times \frac{5}{18} \right) \times (1 - (congestion level - 1) \times 0.25)} + t$$

... .. (eqn. 3)

In the above formula,

the speed limit is multiplied by $\frac{5}{18}$ to convert the speed from kilometers per hour (km/h) to meters per second (m/s),

the factor $(1 - (\text{congestion level} - 1) \times 0.25)$ adjusts the effective speed based on congestion, where each increase in congestion level reduces the speed by 25%. For example, at congestion level 1 (free flow), the factor is 1, meaning no reduction in speed; at level 4 (heavy congestion), the speed is reduced to 25% of its original value,

and t represents the delay caused by traffic signals, calculated as $t = n \times 30 \text{ secs}$, where n is the number of traffic signals encountered and assuming an average delay of 30 seconds per signal.

Graph Construction

To accurately perform the Least Cost Path Analysis (LCPA), we needed to create a detailed graph representation of the road network. This involved segmenting the linestrings, assigning mean congestion values, and finally constructing the graph where the travel cost, represented by edge weights, could be effectively calculated.

Segmenting the Line strings

Initially, each road segment in the network was represented by a single line string. However, to improve the accuracy of our analysis, we decided to segment these line strings further. This segmentation was crucial because the shortest path-finding algorithm operates by evaluating paths between nodes. Long, unsegmented linestrings would have resulted in fewer nodes and, consequently, a less granular analysis. By segmenting each road linestring into smaller segments

of 50 meters, we increased the number of nodes in the graph, thereby enhancing the precision of the pathfinding process.

The process began by converting the road network into a simple feature (sf) object. We then applied a segmentation function to break down each linestring into smaller segments, each not exceeding 50 meters. After segmentation, we added a unique `road\_id` to each segment for identification purposes. These segmented linestrings were then transformed into the projected CRS "EPSG:2022" to ensure accurate distance measurements.

Integrating the Cost Factors: Congestion, Speed Limits, and Traffic Signal Delays

Before constructing the graph, it was essential to incorporate the cost factors being considered: congestion levels, speed limits, and traffic signal delays.

Assigning Mean Congestion Values

We began by assigning mean congestion values to each segmented road linestring. This was done by matching the points along the road segments with the nearest congestion data points. These congestion values were then averaged for each segment, ensuring that each linestring accurately reflected the overall traffic conditions along that portion of the road.

Incorporating Speed Limits

Next, we integrated speed limit data into the road segments. Each segmented linestring was joined with the corresponding speed limit data, ensuring that the legal speed constraints were reflected in the cost metric for each road segment. This step was crucial for accurately calculating the travel time, as it directly affects the speed at which vehicles can traverse each segment.

Traffic Signal Delays

Lastly, we accounted for delays caused by traffic signals. Each road segment was checked for its proximity to traffic signals within a 10-meter radius. If a segment was within this proximity,

a delay of 30 seconds was added to the travel time for that segment. This adjustment provided a more realistic estimate of the total travel time, considering potential stops at intersections.

Constructing the Graph

With the congestion values, speed limits, and traffic signal delays integrated, we proceeded to construct the graph using the `sfnetworks` package in R.

- Edges: Each segmented road line became an edge in the graph, with the travel time cost metric assigned as the weight.
- Nodes: The intersections and endpoints of the segmented linestrings served as the nodes.

The graph was built as an undirected network, with edge weights calculated using the cost formula defined in eqn.3. This weighted graph was then used to determine the most efficient routes within the Toronto Centre Ward.



Figure 4. Road Network Graph



Figure 5. Road Network with assigned weights

Determining the Least Cost Path

To determine the least cost path, we used the ``st_network_paths`` function in R, which implements Dijkstra's algorithm. Dijkstra's algorithm is a well-established method for finding the shortest path between nodes in a graph. It works by iteratively exploring the graph, starting from the source node, and always expanding the least costly path until it reaches the destination node.

In our analysis, this function calculated the optimal route by minimizing the total travel time between the start and end points. Given any start and end points (latitude and longitude), we first identified the nearest nodes in the network. Then, using Dijkstra's algorithm, the most efficient path in terms of travel time was found by considering the weights assigned to the edges, calculated based on the cost metric formulated in eqn. 3.

Forecasting Congestion Values

Following the initial analysis of current traffic conditions, the next step was to forecast future congestion levels and conduct a Least Cost Path Analysis (LCPA) using these predictions. This process involved preparing the data, performing time-series analysis, and finally applying the LCPA to the forecasted data.

Data Preparation and Time Series Formation

To prepare the data for forecasting, we processed multiple raster layers, each representing congestion levels at different times over a week. Due to computational limitations, we selected one traffic congestion raster captured during peak traffic hour (between 8 to 10 am) from each day of the week. These rasters were converted into spatial data and matched with the segmented road network, just as we did in the initial analysis. By doing this for each day, we created a time series for each road segment, showing how congestion levels changed over the seven days.

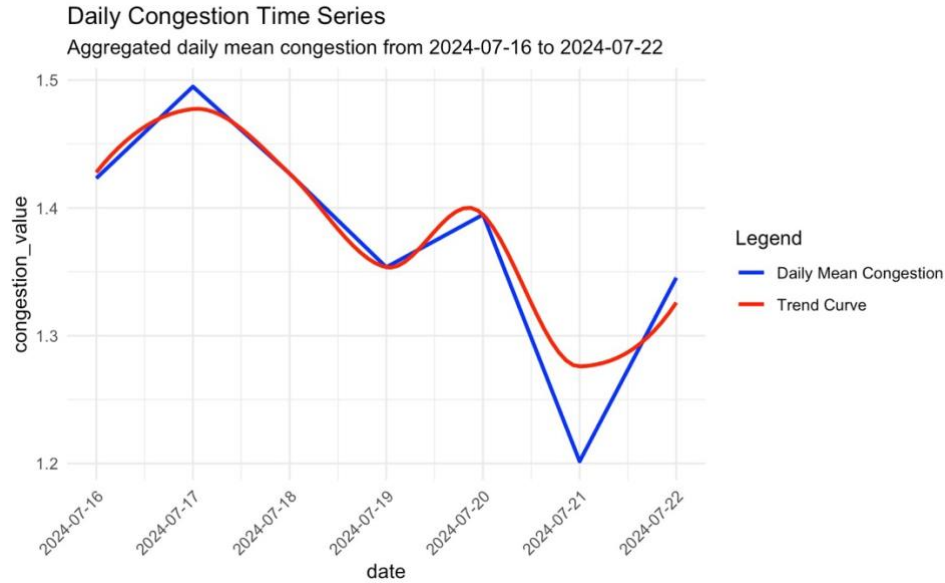


Figure 6: Daily Congestion Time Series

This plot illustrates the average congestion values aggregated from nearly 2,000 road segments in the Toronto Centre Ward, spanning from July 16, 2024, to July 22, 2024. While congestion data is available for each individual road segment, the average values are presented here to provide a clearer view of the overall trend. The plot shows daily fluctuations in congestion, with a peak observed on July 17 which was Wednesday, followed by a significant decrease on July 21 (Sunday). This average trend provides a simplified yet insightful representation of how congestion levels vary over the week in the area.

Forecasting Congestion Levels and Implementing LCPA

With the time series data ready, we forecasted the congestion levels for the eighth day using an ARIMA model. ARIMA is a common method for time-series forecasting and is well-suited for predicting future values based on past data. We trained the model on the seven days of congestion data, which allowed us to predict the expected congestion for each road segment on the following day.

Using the forecasted congestion values for the next day, we updated the weights in the graph network, keeping speed limits and traffic signal locations constant. With these updated weights, we performed the Least Cost Path Analysis (LCPA) just as we did previously and determined the most efficient route based on the anticipated traffic conditions.

Results and Discussion

In this section, we present the outcomes of the Least Cost Path Analysis (LCPA) and the analysis based on the forecasted congestion values for the following day. The results are illustrated through a series of figures that highlight the impact of incorporating congestion as a cost factor in route optimization within the Toronto Centre Ward.

LCPA Without Congestion as a Cost

First, we conducted the LCPA without considering congestion as a cost factor. In this scenario, the paths were determined based solely on the shortest distance, considering only speed limits and traffic signal delays.

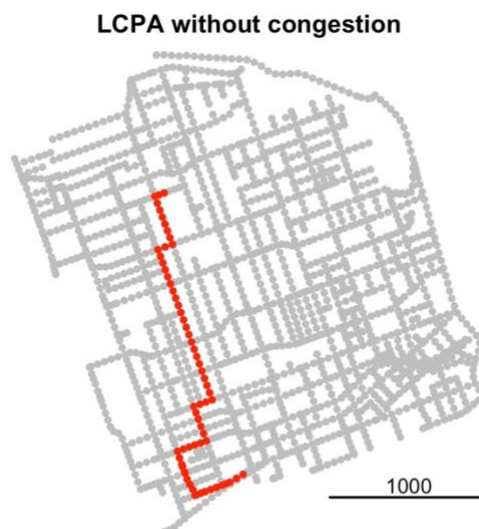


Figure 7: Road network with the least cost path highlighted, excluding congestion as a cost factor.

As shown in Figure 7, the resulting path follows the most efficient path between the start and end points only based on distance, presence of traffic signals and speed limit of the roads. This approach does not account for traffic congestion, which may result in longer travel times during peak hours.

LCPA With Congestion as a Cost

Next, we performed the LCPA by including congestion as a cost factor. This analysis considered the impact of traffic congestion on travel time, leading to the identification of routes that avoid highly congested areas.

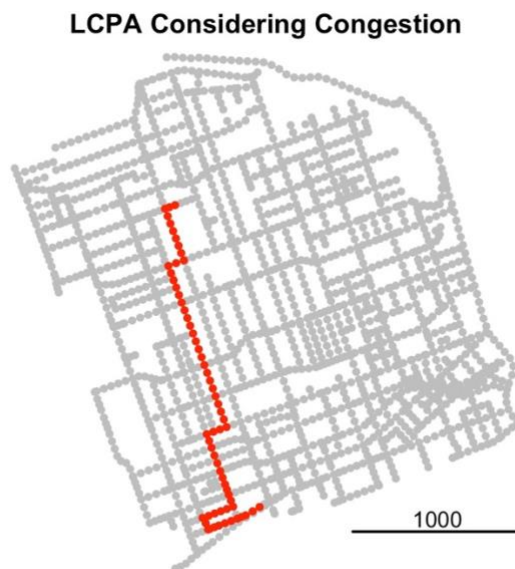


Figure 8: Road network with the least cost path highlighted, including congestion as a cost factor.

In Figure 8, we can see an alternative path to what was seen in Figure 7. The path identified here avoids congested areas, potentially resulting in a longer route in terms of distance but with a

reduced overall travel time. This demonstrates the effectiveness of including congestion in the cost calculation for more accurate and efficient route planning.

Forecasted Congestion and LCPA for the 8th Day

Using the forecasted congestion data for the 8th day, we updated the road network and performed LCPA to identify the most efficient route based on predicted traffic conditions.

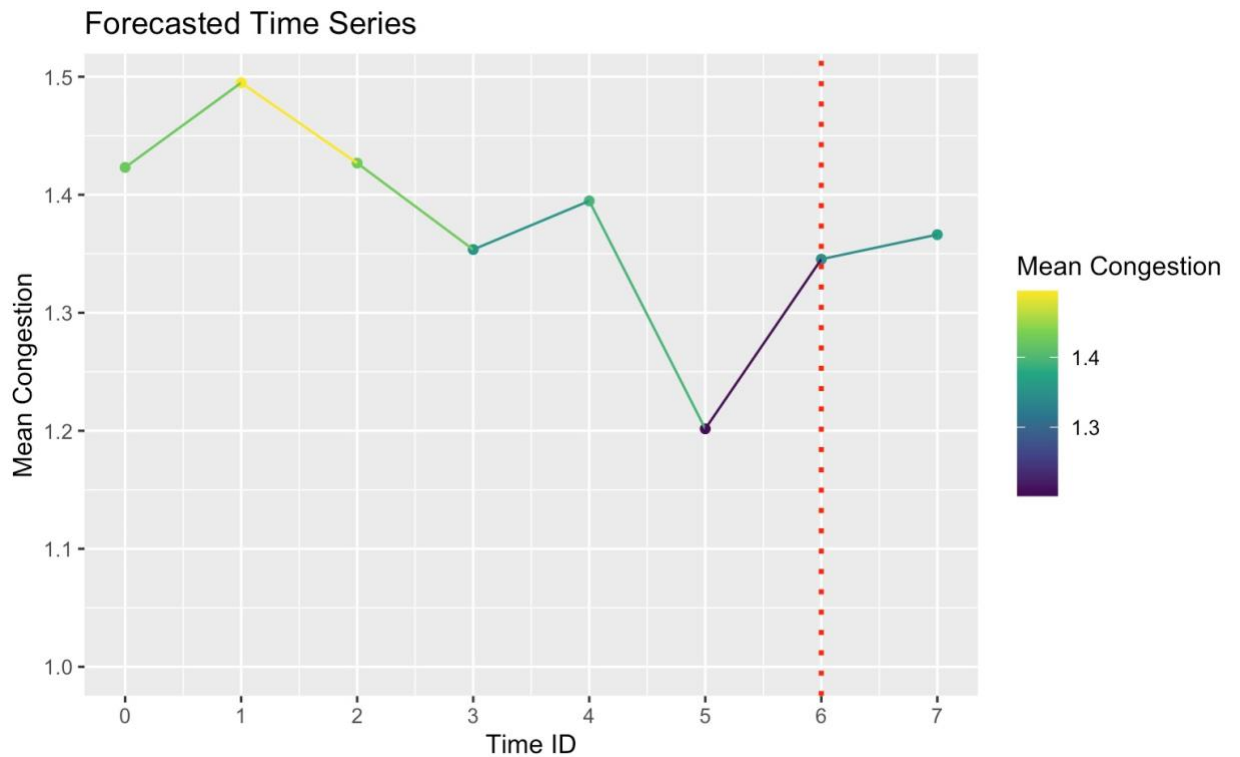


Figure 9: Average Congestion Values Forecasted for the next day

The plot in Figure 9 illustrates the forecasted mean congestion value for July 23, 2024 (Time ID 7), which is derived from the previous seven days of data, spanning from July 16, 2024 (Time ID 0) to July 22, 2024 (Time ID 6). The forecasted value is the average of the predicted congestion levels across all road segments in the Toronto Centre Ward. The average forecasted

congestion value for Tuesday, July 23, 2024, shows a return to a level similar to that observed on the previous Tuesday, July 16, 2024 (Time ID 0).

The forecasted congestion values were then used to update the weights of the graph. These updated weights can be used to perform LCPA as mentioned in previous section. This will allow us to determine the most efficient path based on forecasted congestion values. Below is an illustration of a path between 2 points identified after updating the weights using forecasted congestion values.

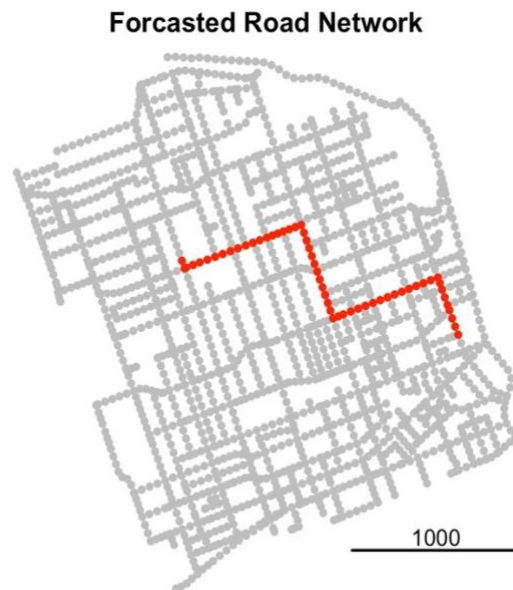


Figure 10: Road network with the least cost path highlighted, based on the forecasted congestion values for the 8th day.

Discussion of Results

This study highlights the importance of incorporating real-time and forecasted congestion data in the Least Cost Path Analysis (LCPA) for urban road networks. The comparison between

routes generated with and without congestion data shows how traffic conditions significantly affect travel efficiency and the choice of routes.

Consideration of Real-Time Congestion:

The LCPA that considers actual, real-time congestion levels (Figure 8) provides a solution that adapts to current traffic conditions. This approach could be valuable in dynamic environments where traffic can change rapidly, such as during rush hour or due to unexpected events like accidents. By using real-time data, the selected route can minimize travel time by avoiding congested areas.

Comparison with LCPA Without Congestion:

The comparison between routes calculated without and with considering congestion (Figures 7 and 8) demonstrates how congestion levels impact the most efficient path in terms of time. While the route without congestion consideration (Figure 7) may appear shorter in distance, it can lead to longer travel times due to unaccounted delays. This shows that neglecting congestion can result in less effective routing, particularly in busy urban areas like Toronto Centre Ward.

Impact of Forecasted Congestion:

The route generated using forecasted congestion data (Figure 10) offers an opportunity to plan routes in advance based on expected traffic conditions. This is especially useful in situations where planning ahead is crucial, such as in logistics or emergency services. By anticipating congestion, it is possible to avoid delays and save time. The time series plot (Figure 9) illustrates how congestion levels vary over time and how this forecasted data can inform the selection of the most efficient route at different times of the day.

Implications and Applications:

The results of this study emphasize the need to include dynamic traffic data in route optimization. Adjusting routes based on both forecasted and real-time congestion data can greatly improve the efficiency of urban transportation. This approach benefits not only individual travelers but also has broader applications in urban planning, logistics, and emergency response, where making timely decisions is critical.

Limitations

While this study offers valuable insights into optimizing route planning in the Toronto Centre Ward, several limitations need to be acknowledged, as they could impact the accuracy and applicability of the results.

Data Accuracy and Granularity:

The accuracy and granularity of the congestion data are significant limitations. The congestion levels were sourced from Google Traffic, which offers a general overview of traffic conditions but may not capture real-time fluctuations with high precision. Since the data was collected at specific intervals, it reflects traffic conditions only at those moments, potentially missing the dynamic nature of traffic throughout the day. This reliance on static data might lead to inaccuracies, especially during unexpected traffic incidents or other events that are not captured in the dataset.

Missing Congestion Values and Imputation:

Another limitation concerns the handling of missing congestion values. In this study, missing data was addressed by averaging the available congestion values along segmented line strings. While this method provides a reasonable estimate, it is not the most robust approach. More

sophisticated imputation techniques could yield more accurate representations of traffic conditions, particularly in areas where data is sparse or missing. Future work could explore these methods to improve the reliability of the analysis.

Limitations in Cost Factors Considered:

This analysis primarily considered congestion levels, speed limits, and traffic signal delays as cost factors for the Least Cost Path Analysis. However, real-world travel times are influenced by many other factors that were not included in this study, such as road conditions, construction zones, temporary road closures, and weather conditions. Additionally, variations in driver behavior, like adherence to speed limits or response to traffic signals, were not considered. The exclusion of these factors could lead to less accurate pathfinding results, especially in complex urban environments.

Simplifications in the Cost Metric:

The cost metric used in this study incorporates speed limits, congestion levels, and traffic signal delays, but it is based on several simplifying assumptions. For example, the assumption that each increase in congestion level reduces speed by 25% is a generalization that may not fully capture the impact of congestion on different roads or at different times. Similarly, the uniform delay of 30 seconds per traffic signal does not account for variations in signal timing or driver behavior. These simplifications could result in discrepancies between the modeled and actual travel times.

Road Segmentation and Point-to-Point Pathfinding:

The road network in this study was segmented into 50-meter line strings to improve the granularity of the analysis. However, this segmentation introduces limitations in determining paths. Specifically, the current analysis is limited to finding paths between nodes, which are the endpoints

of the segmented lines. This means that we cannot select an arbitrary point on a road segment and find the exact path from that point to a destination. Instead, the analysis snaps the start and end points to the nearest nodes, which may not accurately represent the actual path a vehicle would take, especially on longer or more complex road segments. Future work could explore methods to allow for more precise point-to-point pathfinding within the network.

Assumptions of Constant Speed Limits and Signal Locations:

The analysis assumed that speed limits and traffic signal locations remained constant over time. While this assumption is reasonable for short-term analysis, it does not account for changes such as temporary speed limit reductions due to construction, new traffic signals, or changes in signal timing. These factors could impact the accuracy of route recommendations, particularly in urban environments where conditions are subject to frequent changes.

Temporal Scope of Forecasting:

The forecasting model in this study was based on seven days of historical data, which may not capture longer-term trends or seasonal variations in traffic patterns. While the ARIMA model is effective for short-term forecasting, it has limitations in predicting traffic behavior over longer periods or in response to significant changes in traffic flow. Additionally, the model does not consider external factors like road closures, adverse weather conditions, or major events, which could greatly affect traffic conditions and the accuracy of the predicted routes.

Scalability to Larger Networks:

The methods applied in this study were effective within the Toronto Centre Ward, but scaling this approach to larger, more complex road networks may present challenges. Processing and analyzing larger networks with higher spatial resolution could significantly increase

computational demands, requiring more advanced methods or optimizations to maintain efficiency and accuracy.

Justification of Current Analysis and Future Direction

This study offers a valuable framework for real-time and predictive traffic management, but there is room for significant improvement. The simplifications and assumptions made were necessary to balance computational feasibility with the need for actionable insights, but we can enhance this analysis with more advanced resources.

With greater computational power and access to more dynamic and comprehensive datasets, we could incorporate additional variables like real-time traffic updates, road conditions, weather changes, and driver behavior. Including these factors would make the analysis more accurate and reflective of real-world conditions, leading to better and more reliable route recommendations.

We could also scale this approach to cover larger and more complex networks, without sacrificing efficiency or accuracy. This would allow us to apply the methodology to broader regions or more densely populated urban areas, providing even more valuable insights into traffic management.

In future work, we should focus on integrating these more detailed and dynamic data sources, refining the cost metric, and using advanced forecasting models. These improvements would make the analysis more robust and applicable, leading to more effective traffic management solutions that are better suited to the demands of dynamic urban environments.

Conclusion

To sum up, this study demonstrates the usefulness of applying Least Cost Path Analysis (LCPA) to improve route planning in urban areas such as the Toronto Centre Ward. By factoring

in traffic congestion, speed limits, and traffic signals, we identified more efficient routes in a congested part of the city. Incorporating predictions of future traffic conditions also showed how planning ahead can lead to better route choices.

However, the research revealed some areas that need improvement. Using static congestion data captured at specific times may not fully reflect the changing nature of traffic. The approach we used to handle missing data and some of the assumptions in our cost calculations may also have introduced inaccuracies. Additionally, the methods we used for road segmentation and pathfinding between specific points could be refined further.

Even with these limitations, the study offers a solid approach to making urban traffic management more efficient. With more computational resources and collection of more detailed and dynamic data, future work can address these issues by including more variables and applying the methods to larger and more complex road networks, which can ultimately lead to more effective solutions for reducing traffic congestion in urban areas.

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